1.

Prove that two covering maps $p_X : \tilde{X} \to X$ and $p_Y : \tilde{Y} \to Y$ induce a covering map $p_X \times p_Y : \tilde{X} \times \tilde{Y} \to X \times Y$.
Proof: Clearly $p_X \times p_Y$ is continuous, and for each $(x_0, y_0) \in X \times Y$, there exists neighborhoods $U \subset X, V \subset Y$ such that $p_X^{-1}(U)$ and $p_Y^{-1}(V)$ are disjoint homeomorphic copies of U, V. Hence $(p_X \times p_Y)^{-1}(U \times V) = p_X^{-1}(U) \times p_Y^{-1}(V)$ are disjoint homeomorphic copies of $U \times V$.

2.

Given a $d-$sheeted covering map $p : \tilde{n}T^2 \to nT^2$, show that $\tilde{n} = 1 + d(n-1)$.
Proof: Calculating Euler characteristics gives

$$2 - 2\tilde{n} = d(2 - 2n) \implies \tilde{n} = 1 + d(n-1).$$

3.

For each $(d, \tilde{n}, n)$ above, construct a covering map.
Solution: Consider $G = \pi_1(nT^2) = \langle a_1, b_1, \cdots, a_n, b_n \mid [a_1, b_1] \cdots [a_n, b_n] = 1 \rangle$, we take a normal subgroup H of index d, by considering the kernel of the epimorphism $\varphi : G \to \mathbb{Z}/d\mathbb{Z}$ induced by $a_1 \mapsto 1$ and $b_i, a_j \mapsto 0, \forall j \neq 1$. Then H induces a $d-$sheeted normal covering $p : \tilde{X} \to nT^2$, and since p is a local homeomorphism, $\tilde{X}$ is still a orientable closed surface. By classification of closed surfaces, and the same Euler characteristic argument, we know $\tilde{X}$ is $\tilde{n}T^2$.

4.

A subset $A \subset X$ is called semi-simply connected, if A is path-connected and $\iota_* : \pi_1(A) \to \pi_1(X)$ is the trivial map. Prove that, given a covering map $p : \tilde{X} \to X$, and any semi-simply connected open set $U \subset X$, $p^{-1}(U)$ consists of disjoint homeomorphic copies of U.
Proof: For each $u \in p^{-1}(U)$, let $\tilde{U}_u = \{v = \tilde{\gamma}(1) : \gamma = p \circ \tilde{\gamma} \text{ is a path in } U, \tilde{\gamma}(0) = u\}$. Then the restriction $p : \tilde{U}_u \to U$ is a homeomorphism, since it has inverse: for $x_0 = p(u)$ and each $x_1 \in U$, take any path γ in U connecting x_0, x_1, and lift to $\tilde{\gamma}$ in $\tilde{U}_u$, we define $q(x_1) = \tilde{\gamma}(1)$. Since U is semi-simply connected, this inverse is well-defined.
Note that if $v \in \tilde{U}_u$ then $\tilde{U}_v = \tilde{U}_u$ by multiplying a path connecting u, v. Hence $p^{-1}(U)$ is the disjoint union of the different $\{\tilde{U}_u\}$, each homeomorphic to U.

5.

A space X is called locally semi-simply connected, if every point has a semi-simply connected neighborhood. Prove that any covering space of a locally semi-simply connected space is still locally semi-simply connected.
Proof: If $p : \tilde{X} \to X$ is a covering map and X is locally semi-simply connected, then for each point $\tilde{x} \in \tilde{X}, x = p(\tilde{x}) \in X$ has a semi-simply connected neighborhood $U \subset X$. By Exercise4 $p^{-1}(U)$ consists of disjoint homeomorphic copies of U, and we consider the component $\tilde{U}$ containing $\tilde{x}$. Any loop in U is nullhomotopic in X, and the homotopy lifts to a nullhomotopy of loops in $\tilde{X}$, hence $\tilde{U}$ is also semi-simply connected.

6.

Prove that any map $f : S^2 \to T^2$ is null-homotopic.
Proof: Consider any covering $p : X \to T^2$, then since S^2 is path-connected, locally path-connected and simply-connected (so $f_*(\pi_1(S^2, y_0))$ is trivial), f lifts to a map $\tilde{f} : S^2 \to X$. Let X be the universal covering $X = \mathbb{R}^2$, then $\mathbb{R}^2$ is contractible, so every map $\tilde{f} : S^2 \to X$ is null-homotopic. Therefore passing down we obtain $f : S^2 \to T^2$ is null-homotopic.

7.

Prove that non-orientable surfaces have orientable surfaces as a 2-sheeted covering space.
Proof: Given any non-orientable surface nP^2, its Euler characteristic is $\chi(nP^2) = 2 - n$ so we show that there is a covering $p : mT^2 \to nP^2$ where $m = n - 1$. Consider $G = \pi_1(nP^2) = \langle a_1, \cdots, a_n \mid a_1^2 \cdots a_n^2 = 1\rangle$ and the normal subgroup H of index 2: H is the kernel of $\varphi : G \to \mathbb{Z}/2$ given by $a_i \mapsto 1$. Then this subgroup gives a 2-sheeted normal covering $p : \tilde{X} \to nP^2$. Since p is a local homeomorphism, $\tilde{X}$ is still a closed surface, and $\chi(\tilde{X}) = 4 - 2n$. Now note that $H = \mathrm{Ker}\varphi$ consists exactly of those orientation preserving loops, which are the loops which can be lifted to a loop in $\tilde{X}$. Hence every loop in $\tilde{X}$ preserves orientation, so $\tilde{X}$ is orientable.
By classification of closed surfaces, $\tilde{X}$ is $(n-1)T^2$.

8.

Prove that if the universal covering of X is homeomorphic to $\mathbb{S}^n$, then $\pi_1(X)$ is finite.
Proof: Since the universal covering $\tilde{X}$ is homeomorphic to S^n, it is compact, hence the discrete fibers are finite, so $\pi_1(X)$ is finite.

9.

Remove the unit circle of xOy plane and the $z-$axis from $\mathbb{R}^3$, to obtain a space X. Prove that the universal covering of X is $\mathbb{R}^3$.
Proof: Parameterizing by polar coordinates, we know $\varphi : \mathbb{S}^1 \times (0, \infty) \times \mathbb{R} \to \mathbb{R}^3 \setminus z, (v, r, z) \mapsto (rv_x, rv_y, z)$ is a homeomorphism. Hence X is homeomorphic to $(\mathbb{S}^1 \times (0, \infty) \times \mathbb{R}) \setminus (\mathbb{S}^1 \times \{1\} \times \{0\})$, i.e. $\mathbb{S}^1 \times (\mathbb{R}^2 \setminus \mathrm{pt})$, which is then homeomorphic to $\mathbb{S}^1 \times \mathbb{S}^1 \times \mathbb{R}$. Hence its universal covering is $\mathbb{R}^3$ (by Exercise1).

10.

Determine all 3-sheeted coverings of $S^1 \vee S^1$, up to isomorphism.
Solution: $\pi_1(S^1 \vee S^1) \cong \langle a, b\rangle$ is the free group of rank 2. Any group action $\varphi : \pi_1(S^1 \vee S^1) \to S_3$ is induced by the map $\{a, b\} \to S_3$. We classify them by the order of $\varphi(a), \varphi(b)$:
Case1: $|\varphi(a)| = |\varphi(b)| = 1$. Then φ is the trivial map so the covering is three disjoint copies of $S^1 \vee S^1$, $\sqcup_{i=1}^3 (S^1 \vee S^1)$.
Case2: $|\varphi(a)| = 1, |\varphi(b)| = 3$. Then suppose $\varphi(b) = (123)$, this gives the covering space of a triangle with one loop attached at each vertex.

Case3: $|\varphi(a)| = 1, |\varphi(b)| = 2$. This action is not transitive, hence gives the disconnected covering space of $(S^1 \vee S^1 \vee S^1) \sqcup (S^1 \vee S^1)$.
Case4: $|\varphi(a)| = |\varphi(b)| = 2$. If the action is transitive, it gives the covering space $S^1 \vee S^1 \vee S^1 \vee S^1$ (oooo shaped). Otherwise it gives $S^1 \cup S^1$ with two pair of points identified.
Case5: $|\varphi(a)| = 2, |\varphi(b)| = 3$. This gives the shape of a triangle with a loop added at vertex 3 and two additional edges from 1 to 2.
Case6: $|\varphi(a)| = |\varphi(b)| = 3$. This gives the union of two triangles with three corresponding vertices identified, or the
Hence there are a total of 7 distinct covering spaces.

11.

Which coverings in the previous problem are normal?
Solution: A covering induced by a group action is normal iff the centralizer of the image of φ is a transitive subgroup.
Hence the normal coverings are case1 (three disjoint $S^1 \vee S^1$), case2 (a triangle with loops at each vertex), and case6 (two triangles with vertices identified).

12.

Calculate the Deck groups of the normal coverings in Problem10.
Solution: The Deck groups are S_3 for case1, A_3 for case2, and A_3 for case6.